

1. Title of AI Application

Multimodal Medical Foundation Models (MMFMs) for Clinical Diagnosis and Reasoning:
A Focus on the Med-Gemini Paradigm.

2. Domain Selection

Domain: Healthcare and Clinical Diagnostics

The healthcare industry represents one of the most critical, data-dense, and rapidly evolving domains for Artificial Intelligence (AI) adoption. Modern healthcare generates massive amounts of heterogeneous patient data on a daily basis. This includes unstructured text from electronic health records (EHRs), high-dimensional medical imaging (X-rays, MRIs, CT scans, whole-slide pathology), complex genomics sequencing, and real-time continuous telemetry from ICU monitors and wearable sensors. The integration of advanced AI technologies promises to fundamentally revolutionize patient care by ensuring accurate diagnoses, personalizing treatment plans, mitigating human error, and optimizing highly strained clinical workflows.

Recent advancements in artificial intelligence have shifted the medical AI landscape from traditional, narrow "single-task" models—such as a computer vision model trained exclusively to detect lung nodules on a CT scan—to Multimodal Medical Foundation Models (MMFMs). These generalist models, exemplified by architectures like Google's Med-Gemini, utilize a unified, end-to-end learning framework to ingest and process text, medical images, genomics data, and time-series data simultaneously. By mirroring how human clinicians holistically review a patient's history, imaging, and labs to arrive at a diagnosis, MMFMs can reason through complex clinical vignettes, synthesize dense radiology reports, and suggest precise treatment pathways.

The adoption of AI in clinical diagnostics offers immense societal and economic benefits. These include drastically reduced diagnostic turnaround times, the early detection of life-threatening conditions (such as aggressive cancers and degenerative neurological

disorders), the alleviation of severe physician burnout, and the democratization of access to expert-level medical reasoning in under-resourced or rural areas. Therefore, Clinical Diagnostics using Multimodal Medical Foundation Models has been selected as the application domain for this report, as it represents the absolute frontier of intelligent, lifesaving technological innovation in modern medicine.

3. Introduction

Artificial Intelligence (AI) has initiated a paradigm shift in healthcare by enabling computational systems to analyze complex medical data, uncover hidden pathological patterns, and assist clinicians in high-stakes decision-making. Historically, medical AI has been highly siloed and modular. A modern hospital might employ one specific algorithm to flag sepsis risks strictly from EHR vital signs, a separate, disconnected computer vision model to triage chest X-rays in the emergency department, and yet another natural language processing (NLP) tool to extract billing codes from clinical discharge notes.

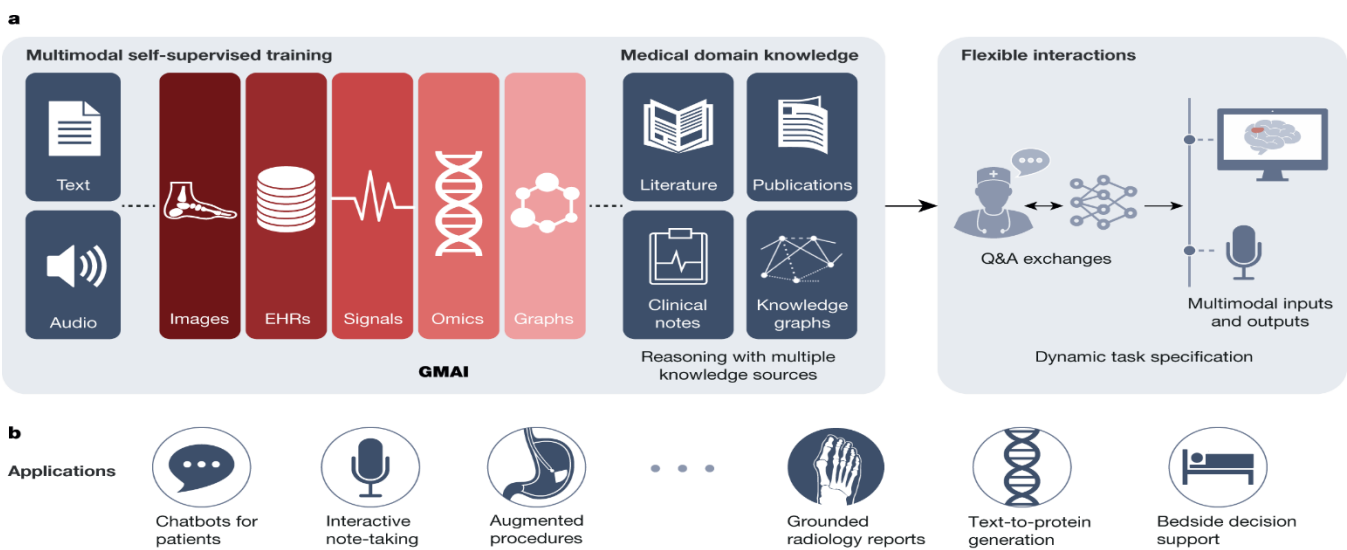
While these modular, single-modality systems have advanced patient care and achieved FDA approvals, they suffer from fundamental architectural and operational limitations. Patients present as whole, complex biological systems, not as isolated, clean datasets. When the diagnostic process is split across disconnected AI modules, critical contextual links are lost. For example, a very subtle, borderline abnormality on a brain MRI might only become clinically significant when combined with a specific, obscure patient symptom documented in a clinical note and a rare genetic marker found in lab results. Traditional modular AI cannot bridge these inter-modal gaps, leading to a fragmented, incomplete understanding of patient health and limiting the system's ability to handle unpredictable real-world clinical scenarios.

To overcome these profound challenges, researchers have developed Multimodal Medical Foundation Models (MMFMs). These systems employ advanced self-supervised learning on massive datasets of diverse, unlabeled, or naturally paired biomedical data (e.g., paired

X-rays and their corresponding text-based radiology reports). MMFMs integrate text, vision, and structured clinical data into a single, shared representational space. This unified architecture enables the AI to perform complex, multi-step clinical reasoning, moving beyond simple pattern recognition to approximate human-like diagnostic deduction.

The latest breakthrough in this space is the development of Med-Gemini, a family of highly capable clinical AI models developed by Google Research and DeepMind. Built upon the robust Gemini architecture and explicitly fine-tuned on vast, specialized medical datasets, Med-Gemini has achieved state-of-the-art accuracy, including scoring 91.1% on the MedQA US Medical Licensing Examination (USMLE) benchmark. It processes multimodal inputs in a single inference pass and uniquely utilizes uncertainty-guided search mechanisms to fetch up-to-date clinical evidence when its confidence is low. This drastically reduces AI hallucinations—a critical, non-negotiable requirement for deployment in medical environments.

This report explores the underlying principles, AI technologies, complex data pipelines, real-world implementations, benefits, limitations, and future scope of Multimodal Medical Foundation Models in healthcare, highlighting how they are transitioning modern medicine from reactive, fragmented care to proactive, precision-driven clinical intelligence.



Regulations: Application approval; validation; audits; community-based challenges; analyses of biases, fairness and diversity

4. Problem Identification

The development of highly reliable, generalist AI for clinical diagnostics faces several intense technical, ethical, and operational challenges that have historically bottlenecked the widespread deployment of autonomous medical systems:

Data Fragmentation and Silos:

In real-world clinical practice, patient data is notoriously scattered across incompatible, legacy IT systems. Medical images are stored in PACS (Picture Archiving and Communication Systems), bloodwork in LIMS (Laboratory Information Management Systems), and physician notes in various proprietary EHRs (Electronic Health Records). Traditional AI models struggle to ingest, synchronize, and synthesize these disparate data types simultaneously, leading to incomplete analyses.

The "Black Box" Problem and Interpretability:

Deep learning models inherently operate as complex neural networks with billions of parameters, making it exceedingly difficult to trace exactly *why* an AI made a specific diagnostic recommendation. In medicine, where life-or-death decisions are made and legal liability is high, interpretability and clinical trust are essential. A system that simply outputs "Cancer: 98% probability" without providing a physiological rationale is unusable by a human doctor.

Domain Shift and Generalization Failures:

Medical AI is highly susceptible to demographic and technological domain shifts. An AI model trained on pristine, high-resolution MRI data from a wealthy, urban research hospital may perform dangerously poorly when deployed in a rural clinic using older, lower-resolution scanning equipment. Furthermore, models trained predominantly on specific ethnicities may misdiagnose underrepresented populations, leading to severe ethical and equity issues.

Missing Modalities and Incomplete Records:

In the real world, patients rarely present with perfectly complete medical records. A patient arriving at an emergency room might have immediate bloodwork and clinical notes, but no relevant historical imaging or genomic sequencing. AI models must be robust enough to reason accurately and degrade gracefully even when expected data modalities are missing.

Physician Burnout and Cognitive Overload:

The sheer volume of medical data generated per patient today far exceeds human cognitive limits. Clinicians spend hours sifting through dozens of pages of EHR data just to prepare for a 15-minute consultation. This leads to severe physician burnout, which in turn drastically increases the risk of cognitive errors, missed diagnoses, and patient harm.

Hallucinations in Generative AI:

While general-purpose Large Language Models (LLMs) are highly articulate, they are prone to confident fabrication, known as "hallucination." In a medical context, an AI hallucinating a nonexistent clinical guideline or suggesting a contraindicated drug dosage is catastrophic. Thus, medical AI requires highly specialized fine-tuning and grounding mechanisms to ensure factual accuracy.

5. Need for AI Solution

The escalating complexity of modern precision medicine makes it virtually impossible for human clinicians to manually synthesize every single data point available for every patient. According to global health studies, medical errors—often stemming from missed information, cognitive overload, or delayed diagnoses—are a leading cause of patient harm and mortality worldwide.

Traditional rule-based Clinical Decision Support Systems (CDSS) have attempted to solve this but have largely failed. These systems rely on rigid, manually programmed logic (e.g., "If heart rate > 100 and temperature > 101, flag for infection"). This approach triggers "alert fatigue"—bombarding doctors with thousands of generic, hard-coded warnings that are routinely ignored because they lack patient-specific nuance. Furthermore, maintaining isolated AI point-solutions requires hospitals to juggle dozens of separate vendor contracts, creating unmanageable IT infrastructure.

A unified, AI-driven Foundation Model provides a comprehensive, intelligent solution:

1. **Holistic Patient View:** By natively understanding both the visual anomaly on a CT scan and the semantic meaning of the patient's complaints in the clinical notes, the AI can generate highly accurate, context-aware differential diagnoses that a single-modality AI would miss.
2. **Workflow Automation:** AI can instantly draft radiology reports, summarize dense patient histories, and highlight urgent cases, giving doctors hours of their day back to focus on direct patient interaction and empathy.
3. **Early Detection:** Advanced computer vision capabilities can detect micro-calcifications, subtle neurological changes, or genomic markers months or years before a human expert might notice them, dramatically improving survival rates.

4. Personalized Medicine: By integrating genomics with clinical histories, AI can predict exactly how a specific patient will respond to a targeted therapy, optimizing the treatment dosage and minimizing the risk of adverse drug reactions.

Table 1: Comparison Between Traditional Medical AI and MMFMs

Feature	Traditional Medical AI (Rule-Based/Single-Task)	Multimodal Medical Foundation Models (MMFMs)
Decision Making	Predefined rules or narrow pattern matching	Context-aware, human-like deductive reasoning
Architecture	Independent, siloed modular algorithms	Unified, End-to-End deep learning framework
Data Handled	Single modality (e.g., only X-rays or only text)	Text, images, EHR, time-series, and genomics combined
Adaptability	Requires full retraining from scratch for new tasks	Adapted via simple prompting or light fine-tuning (Zero-shot)
Handling Edge Cases	Poor; fails when data falls outside strict parameters	Better generalization through massive pre-training
Workflow Impact	Adds separate, disjointed steps to clinician workflow	Consolidates tasks into a single, cohesive platform

6. AI Technologies Used

The architecture of Multimodal Medical Foundation Models relies on an intricate interplay of cutting-edge AI technologies. By moving beyond basic machine learning into advanced generative and self-supervised paradigms, these systems mimic the cognitive processes of medical professionals.

6.1 Computer Vision (Large Vision Models)

Medical imaging requires a level of precision far exceeding standard computer vision tasks. MMFMs utilize deep Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to process high-dimensional imaging. This includes 2D X-rays, 3D volumetric scans (like MRIs and CTs), and billion-pixel whole-slide digital pathology images. These models are responsible for feature extraction, precise tumor segmentation, and the detection of subtle anatomical anomalies that are invisible to the naked eye.

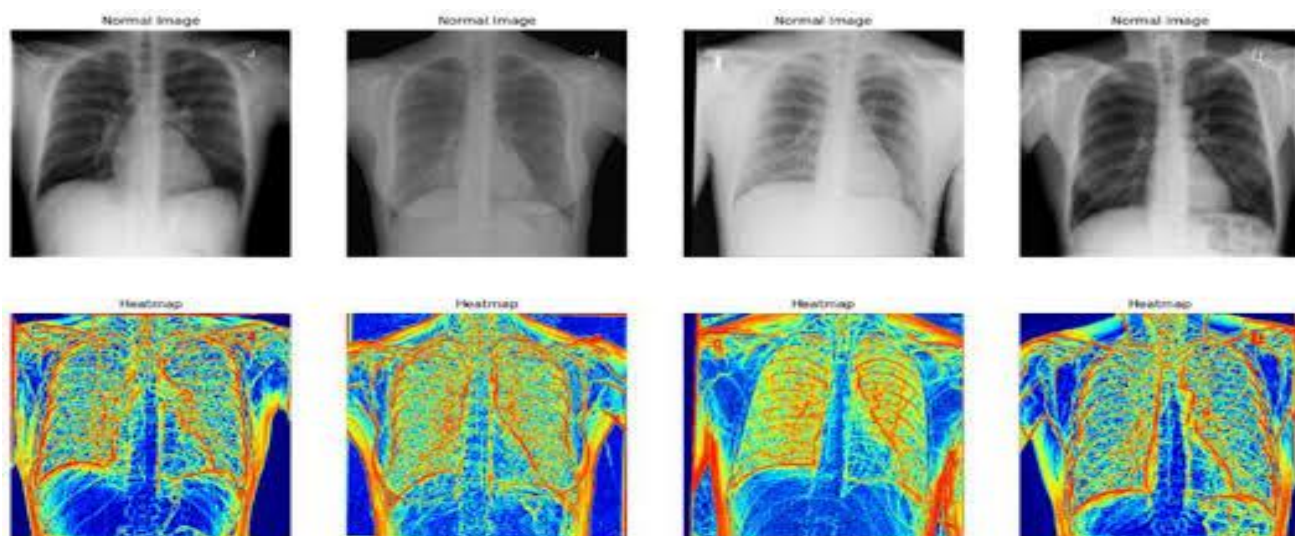
6.2 Large Language Models (LLMs)

Large Language Models serve as the core reasoning and communication engine. They process unstructured clinical text, medical journals, and raw EHR data. Fine-tuned medical LLMs can parse nuanced symptoms, understand incredibly complex medical terminology and abbreviations, and generate human-readable differential diagnoses, discharge summaries, and step-by-step treatment plans.

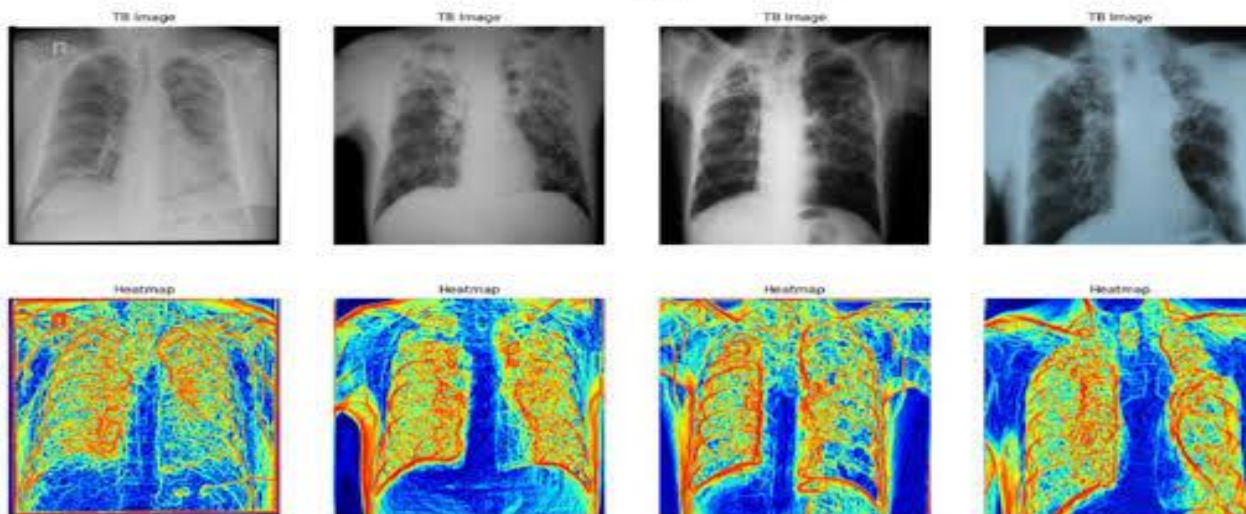
6.3 Transformer Networks and Self-Attention

Transformer Networks are the backbone of modern MMFMs. Unlike older recurrent networks, Transformers allow the model to process massive sequences of data by weighting the importance of different inputs simultaneously. The Self-Attention mechanism is critical for cross-modal fusion; it allows the model to "attend" to a specific pixel (e.g., a shadow) in an X-ray while simultaneously "attending" to a specific word (e.g., "cough") in the patient's clinical note.

This mechanism ensures the model focuses on the most clinically relevant information across all data modalities when making a diagnosis.



(a) Normal



(b) Tuberculosis

6.4 Contrastive Learning (Self-Supervised Learning)

Because manually labeling medical data by highly paid doctors is prohibitively expensive and slow, MMFMs heavily utilize Self-Supervised Learning. Contrastive learning pairs naturally occurring data, such as a routine X-ray image and the text-based radiology report written by the doctor on that same day. The model is mathematically trained to minimize the distance between matching pairs and maximize the distance between non-matching pairs in the latent space. This allows the AI to learn complex human anatomy and pathology entirely without manual human labeling.

6.5 Retrieval-Augmented Generation (Uncertainty-Guided Search)

To combat the fatal flaw of LLM hallucinations, advanced models like Med-Gemini employ Retrieval-Augmented Generation (RAG) and uncertainty-guided search. When the model encounters a highly complex clinical scenario where its internal confidence threshold drops, it dynamically queries external, verified medical databases (like PubMed or institutional clinical guidelines) in real-time. It retrieves the latest, peer-reviewed evidence and incorporates it into its final reasoning, ensuring the output is factually grounded.

Table 2: AI Technologies Used in MMFMs

AI Technology	Primary Function	Application in Healthcare
Computer Vision (ViTs)	Spatial & visual perception	Detects tumors, fractures, and histological anomalies
Large Language Models	Semantic understanding	Parses clinical notes, generates reports and discharge summaries
Transformer Networks	Cross-modal contextualization	Links visual anomalies in scans to text-based symptoms
Contrastive Learning	Self-supervised pre-training	Learns medical concepts from unlabeled, paired hospital data
RAG / Guided Search	Factual grounding	Retrieves real-time clinical guidelines to prevent hallucinations

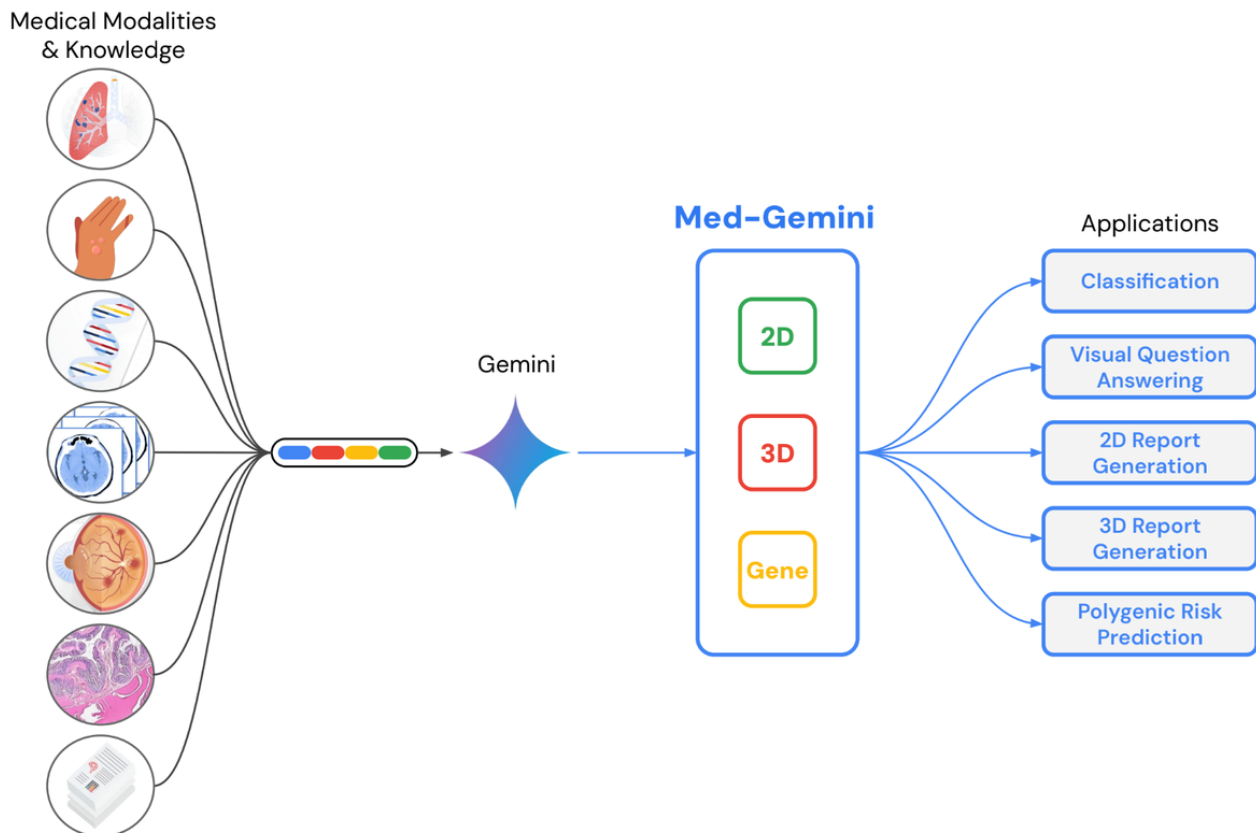
7. Working Mechanism of the Proposed AI System

The operational pipeline of a Multimodal Medical Foundation Model (such as Med-Gemini) functions through a unified, end-to-end ingestion and reasoning process. Unlike older systems that require manual hand-offs between software, the VLA (Vision-Language-Action) equivalent in medicine handles the entire cognitive load continuously.

1. **Multimodal Input Layer:** The model simultaneously receives diverse streams of patient data. For example, a clinician might input a digital chest X-ray, a real-time string of vital signs (heart rate, SpO2, blood pressure), and a natural language text prompt: *"Patient is a 65-year-old male presenting with acute shortness of breath and a 30-year history of smoking. Analyze this X-ray, consider his vitals, and suggest next diagnostic steps."*
2. **Modality-Specific Encoding:**
 - The Vision Encoder processes the X-ray, utilizing Vision Transformers to extract high-level spatial features (e.g., identifying a localized opacity in the lower left pulmonary lobe).
 - The Text/Sequence Encoder processes the clinical prompt and vital signs, contextualizing the semantic weight of a "history of smoking" combined with "shortness of breath" and a dropping SpO2 level.
3. **Cross-Modal Fusion:** The extracted, high-dimensional representations from both encoders are projected into a shared latent space. Through complex cross-attention layers, the model mathematically links the visual opacity found in the scan directly to the semantic concepts of pneumonia, chronic obstructive pulmonary disease (COPD), or lung cancer.
4. **Clinical Reasoning & Search:** The language model acts as the central reasoning engine, processing the fused data. If the differential diagnosis is highly complex or ambiguous, the *Uncertainty-Guided Search* module is automatically triggered. The

AI pulls the latest clinical guidelines for treating suspected pulmonary masses in patients with the exact demographic and vital profile presented.

5. **Output Generation:** Finally, the model synthesizes a comprehensive response. It outputs a prioritized list of differential diagnoses, visually highlights the region of interest on the X-ray via a saliency map (providing vital interpretability for the doctor), and generates a drafted, standardized radiology report ready for physician review and sign-off.



8. Data Sources and Processing

The sheer performance of a Multimodal Medical Foundation Model depends entirely on the quality, diversity, and volume of the data used for pre-training and fine-tuning. These datasets must be meticulously processed to preserve patient privacy (HIPAA/GDPR compliance) while maintaining clinical integrity.

8.1 Data Sources

The proposed MMFM system collects and learns from the following critical sources:

- **PACS (Picture Archiving and Communication Systems):** The primary source of DICOM-format medical images. This includes high-resolution X-rays, MRIs, CT scans, and Ultrasounds. PACS data provides the morphological and structural ground truth of patient disease.
- **EHR (Electronic Health Records):** The central nervous system of hospital data. EHRs contain unstructured physician progress notes, detailed discharge summaries, structured laboratory results, demographic information, and historical medication lists.
- **Genomics and Multi-omics Databases:** Deep biological data, including DNA sequencing, RNA expression profiles, and specific mutation panels (like BRCA1/2). This data is absolutely crucial for precision oncology and predicting responses to specific chemotherapies.
- **Time-Series and Telemetry Data:** Continuous, high-frequency data from ICU monitors, including ECG waveforms, ventilator pressure readings, and modern wearable health devices (smartwatches tracking atrial fibrillation).

8.2 Data Processing Pipeline

Before a single neural network weight is updated, the multimodal data undergoes rigorous preprocessing:

- **De-identification and Anonymization:** All Protected Health Information (PHI)—such as names, dates of birth, and addresses—is rigorously stripped from both text and image metadata using specialized NLP models to ensure strict legal compliance.
- **Data Synchronization & Alignment:** Multimodal data must be aligned temporally. The system must ensure that a specific blood test logically corresponds to the correct MRI scan on the patient's clinical timeline, avoiding dangerous data mismatches.
- **Pre-training (Self-Supervised):** The model is pre-trained on massive amounts of unlabeled, paired medical data (e.g., millions of X-rays and their text reports) to learn general biomedical representations and anatomy.
- **Clinical Fine-Tuning (Supervised):** Following pre-training, the model undergoes supervised fine-tuning on high-quality, doctor-annotated datasets (such as the MedQA benchmark) to refine its step-by-step clinical reasoning and ensure its outputs align with actual medical protocols.

Table 3: Data Sources Used in Medical Foundation Models

Data Source	Type of Data	Clinical Purpose
PACS	DICOM Images (MRI, CT, X-Ray)	Anatomical visualization, tumor detection, structural anomalies
EHR	Unstructured Text & Lab Values	Patient history, symptom tracking, medication interactions
Genomics	DNA/RNA Sequencing Panels	Precision medicine, targeted oncology, inherited risk profiling
Telemetry	Time-Series (ECG, Vitals)	Real-time monitoring, predicting acute ICU deterioration

9. Real-World Implementation and Case Studies

The rapid advancement of AI has accelerated the deployment of foundation models across leading healthcare institutions and tech companies. These real-world implementations demonstrate the practical, life-saving feasibility of MMFMs.

9.1 Google DeepMind: Med-Gemini

Released as a major breakthrough, Med-Gemini is a family of highly capable, clinically native multimodal models. Unlike general-purpose LLMs, it was fine-tuned specifically on medical datasets (like MedQA-R and MedQA-RS). Med-Gemini possesses the unique capability to seamlessly integrate 3D medical imaging, such as highly complex whole-slide histopathology and CTs, with long-context EHR text. Leading research institutions are utilizing it for complex genomic risk encoding, automated clinical trial matching, and providing second opinions on rare diseases where human specialists are unavailable.

9.2 OpenAI: GPT-4 for Healthcare (General Purpose Adaptations)

While not exclusively a medical model from the ground up, GPT-4 (and its subsequent iterations) has seen massive integration via enterprise healthcare platforms, most notably through partnerships with EHR giants like Epic Systems. It is primarily utilized as an "ambient clinical scribe." The AI securely listens to doctor-patient conversations in the examination room, extracts the medical intent, and automatically drafts highly structured clinical notes. This implementation is currently saving physicians several hours of administrative work each day, directly combating burnout.

9.3 Mayo Clinic & Diagnostic Generative AI

Top-tier research hospitals like the Mayo Clinic are pioneering the use of MMFMs to predict acute patient deterioration in the Intensive Care Unit (ICU). By continuously feeding chest X-rays, real-time ventilator waveforms, and dynamic nursing notes into a unified model.

9.4 PathAI / Paige (Digital Pathology)

Companies like PathAI and Paige have deployed highly specialized foundation models focused on digital pathology. By analyzing billion-pixel images of biopsied tissue, these models identify micro-metastases of breast and prostate cancer that human pathologists, suffering from eye fatigue, might miss. They also quantify specific biomarkers to determine exactly which immunotherapy drug a patient should receive.

Table 4: Comparison of Real-World Medical AI Implementations

Platform	Primary AI Technology	Core Modalities Used	Key Clinical Strength
Google Med-Gemini	Multimodal Foundation Model	Vision, Text, Genomics, EHR	Diagnostic accuracy, Uncertainty-guided clinical search
OpenAI GPT-4 (Health)	Large Language Model	Text, Audio (Ambient scribe)	Workflow automation, Clinical documentation and summaries
Mayo Clinic ICU AI	Sensor Fusion + Deep Learning	Time-series, X-rays, EHR	Real-time predictive analytics, Early deterioration warning
PathAI / Paige	Vision Transformers	Digital Pathology (Slides)	Cancer detection, Biomarker quantification at microscopic level

10. Advantages, Limitations, and Future Scope

10.1 Advantages

The adoption of Multimodal Medical Foundation Models offers several profound advantages over traditional healthcare IT and rule-based systems:

- **Holistic Decision Making:** By combining disparate patient data streams (images, text, genomics) into one comprehensive clinical picture, MMFMs catch diagnoses that siloed analysis misses.
- **Democratization of Expertise:** These models provide general practitioners in rural or under-resourced areas with diagnostic insights comparable to board-certified subspecialists, dramatically improving global health equity.
- **Reduced Administrative Burden:** By automating note-taking, report generation, and billing code extraction, AI allows doctors to focus on empathy and direct patient care.
- **Label-Efficient Learning:** The use of self-supervised learning on raw hospital data drastically reduces the need for expensive, time-consuming datasets annotated by human doctors.

10.2 Limitations

Despite their immense potential, MMFMs face severe technical and regulatory hurdles:

- **Compute and Infrastructure Costs:** Training and running real-time inference on billion-parameter models requires massive, highly expensive GPU clusters. This effectively prices out smaller, independent health systems from running models locally.
- **Algorithmic Bias and Domain Shift:** If a model is trained predominantly on data from specific demographics (e.g., wealthy populations in the Global North), it may misdiagnose underrepresented populations, exacerbating healthcare inequalities.

- **The Interpretability Challenge:** Physicians are legally and ethically responsible for their diagnoses. The "black box" nature of deep learning makes it difficult to justify AI-assisted decisions in medical malpractice scenarios.
- **Data Privacy & Security:** Aggregating massive amounts of multimodal patient data raises severe privacy concerns. Centralized databases are prime targets for cyber-attacks, necessitating flawless encryption and governance.

10.3 Future Scope

The future of autonomous and AI-assisted medicine will be driven by the continuous refinement of foundation models and infrastructure:

- **Federated Learning:** To solve privacy issues, future MMFMs will utilize Federated Learning. This allows the AI model to train across thousands of hospitals globally without ever moving or centralizing the sensitive patient data from the local hospital servers.
- **Embodied Clinical Agents:** We will see a shift from passive AI to active AI agents. These systems will not only suggest a diagnosis but actively interact with hospital software to automatically order the necessary follow-up labs, schedule MRIs, and manage smart drug-delivery IV pumps.
- **Digital Twins and Synthetic Data:** AI will generate highly accurate "digital twins" of patients using synthetic data. Researchers will be able to simulate massive clinical trials computationally in days rather than years, vastly accelerating drug discovery for rare diseases.
- **Explainable AI (XAI):** Significant research will focus on forcing models to show their exact mathematical "work," generating causal maps that prove to the physician exactly why a specific diagnosis was rendered.

Table 5: Advantages, Limitations, and Future Scope of MMFMs

Category	Highlights
Advantages	Holistic reasoning, democratization of specialist expertise, reduced doctor burnout, label-efficient self-learning.
Limitations	Massive computational costs, algorithmic bias, lack of clinical interpretability, severe data privacy risks.
Future Scope	Federated Learning, Embodied Clinical Agents, Digital Twins for drug discovery, Explainable AI (XAI).

11. Conclusion

The integration of Artificial Intelligence into healthcare has evolved dramatically from isolated, single-task algorithms into robust, highly intelligent Multimodal Medical Foundation Models (MMFMs). This report examined the application of these systems in clinical diagnostics, highlighting how the convergence of Computer Vision, Large Language Models, Transformer architectures, and Self-Supervised Learning has birthed clinical systems capable of holistic patient reasoning, exemplified by state-of-the-art models like Med-Gemini.

By successfully breaking down the historical data silos between medical imaging, electronic health records, continuous telemetry, and genomics, MMFMs closely mirror the complex, multi-step deductive reasoning of human physicians. Real-world implementations across platforms like Google DeepMind, OpenAI, and leading research hospitals are already demonstrating massive potential to reduce diagnostic errors, alleviate catastrophic physician burnout, and pioneer a new era of precision medicine.

However, realizing the full, transformative potential of these systems requires the medical and tech communities to overcome significant hurdles. Issues regarding exorbitant computational costs, clinical interpretability, demographic bias, and strict patient privacy laws remain active battlegrounds. As advanced technologies like Federated Learning, Retrieval-Augmented Generation, and Explainable AI mature, MMFMs will undoubtedly overcome these barriers. Ultimately, Multimodal Medical Foundation Models represent a monumental milestone in intelligent healthcare, poised to become the foundational infrastructure of modern medicine, fundamentally transforming how patient care is delivered, managed, and optimized on a global scale.

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8. The adaptation of models for specialized tasks is covered in "Med-Flamingo: a Multimodal Medical Few-shot Learner".
9. Research detailing the creation of conversational assistants is documented in "LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day".
10. The expansion of visual processing capabilities is explored in "Medical Large Vision Language Models with Multi-Image Visual Ability".